

Core Competencies

- Data Platform Architecture & Solution Design
- Cloud Data Engineering (AWS, Multi-Cloud)
- Big Data Processing (Apache Spark, Hadoop Ecosystem)
- ETL/ELT Pipeline Engineering (Batch & Streaming)
- Data Lakehouse Architecture (Iceberg, S3)
- Data Reliability, Observability & Governance
- Real-Time Data Streaming (Apache Kafka)
- Performance Optimization & Cost Engineering
- Data Modeling & Analytics Enablement
- Cross-functional Stakeholder & Solution Consulting

Certifications

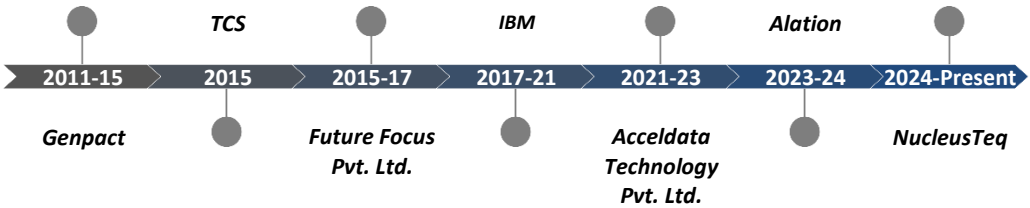
-  AWS Certified Solutions Architect – Associate
-  AWS Certified Machine Learning – Speciality

Major Highlights Across Career

Profile Summary

- Senior Data Engineering professional with **14+ years of experience**, including **7+ years in modern data engineering**, specializing in designing **cloud-native, scalable data platforms** and distributed data processing systems.
- Proven expertise in building and optimizing **high-volume ETL/ELT pipelines (batch & real-time)** using Apache Spark, Airflow, and Kafka, enabling **analytics, BI, and AI-driven decision systems**.
- Strong foundation in **data platform architecture and solution design**, with growing alignment toward **Solutions Architect responsibilities**, including system design, technology selection, and scalable data ecosystem planning.
- Hands-on experience in implementing **data reliability, quality, and governance frameworks**, ensuring high data trust through SLA-driven monitoring, anomaly detection & secure access controls.
- Delivered **high-impact transformations**, including **zero-downtime, zero-data-loss migration across big data platforms**, and development of robust data layers supporting enterprise-scale analytics and product use cases.

Career Timeline



Technical Skills

- Big Data & Distributed Systems:** Hadoop (HDFS, YARN, Hive, HBase, Impala, ZooKeeper), Apache Spark (Spark SQL, AQE, Shuffle Optimization), Apache Iceberg, Trino, Presto
- Streaming & Event Processing:** Apache Kafka
- Cloud Platforms:** AWS (EMR, S3, EC2, IAM), Microsoft Azure (HDInsight), Google Cloud Platform (Dataproc, BigQuery)
- Data Governance & Security:** RBAC, Apache Ranger, Kerberos, Data Quality Frameworks, Monitoring, Observability
- Databases:** PostgreSQL (EDB), Oracle, MySQL, Hive Metastore
- Programming & Scripting:** Python, SQL, Scala, PySpark, Unix Shell Scripting
- Orchestration:** Apache Airflow
- DevOps & Tools:** Git, CI/CD (Jenkins/GitHub Actions – exposure), Docker
- Operating Systems:** Linux, Windows

Work Experience

NucleusTeq | Senior Data Platform Engineer | May 2024 – Present

- Architecting and delivering **scalable batch and real-time data pipelines** using Apache Spark and Apache Airflow on AWS EMR, processing **terabyte-scale datasets** to support enterprise analytics and data-driven applications.



Improved **distributed data pipeline performance (Spark-based ETL/ELT systems)** by reducing job execution time from **2–3 hours to 20–30 minutes (70–85% improvement)** through advanced optimization techniques including partition tuning, AQE, and shuffle optimization.



Delivered a **zero-downtime, zero-data-loss migration** of a **multi-TB scale big data platform**, ensured seamless transition of enterprise workloads without business disruption or replication overhead.



Architected and supported **scalable data platforms processing 10+ TB/day workloads**, enabled high-volume batch and real-time analytics consumption for enterprise BI and downstream AI/ML systems.



Implemented **data reliability and observability frameworks** (freshness, completeness, schema validation, anomaly detection), resulted in a **30–40% reduction in production incidents and improved SLA compliance** across data pipelines.



Designed optimized **analytics-ready data layers using Spark, Trino, and Iceberg**, improved query performance and reduced **data retrieval latency by 40–60%**, enabled faster business decision-making.



Optimized AWS-based data processing infrastructure (EMR, S3, Spark workloads), achieved **25–35% reduction in cloud compute costs** through resource tuning, workload balancing, and execution efficiency improvements.

- Engineering **high-performance analytical data layers** leveraging Trino and Apache Iceberg, enabling **self-service analytics, BI reporting, and product intelligence use cases** across business teams.
- Managing and orchestrating **multiple production-grade workflow**, ensuring seamless data availability for **cross-functional stakeholders with both daily and streaming data requirements**.
- Driving implementation of **automated data quality and reliability frameworks**, incorporating freshness checks, completeness validation, and anomaly detection to **enhance data trust and reduce production incidents**.
- Optimizing distributed data processing workloads through **advanced Spark tuning techniques (memory optimization, partitioning, shuffle tuning, AQE)**, improving performance and **reducing cloud infrastructure costs**.
- Owning **end-to-end data pipeline lifecycle management**, including monitoring, alerting, incident resolution, SLA adherence, and **reducing MTTR through proactive observability practices**.
- Enforcing **enterprise-grade data governance and security controls**, including RBAC and compliant access mechanisms, while contributing to **scalable data platform architecture** supporting analytics, machine learning, and product innovation initiatives.

Alation | Staff Technical Engineer | May 2023 – Feb 2024

- Designed and developed **scalable data pipelines for an enterprise-grade Data Catalog platform**, enabled metadata ingestion, enrichment, lineage tracking, and **usage intelligence across global data ecosystems**.
- Built and enhanced **metadata ingestion frameworks** to integrate diverse enterprise data sources, improved **data discoverability, accessibility, and trusted consumption**.
- Developed advanced **data transformation and enrichment logic**, strengthened metadata standardization, lineage accuracy, and **ownership mapping across datasets**.
- Translated business and product requirements into **scalable data models and analytics-ready schemas**, supported **data discovery insights and platform adoption metrics**.
- Implemented **automated data validation, schema enforcement, and consistency checks**, ensured high-quality metadata and reliability of downstream data processes.
- Tuned Spark and SQL-based processing jobs through **execution plan optimization, partitioning strategies, and performance engineering**, enhanced efficiency and platform scalability.

Acceldata Technology Pvt. Ltd. | Solutions Engineer | Apr 2021 – May 2023

- Contributed to the development of a **data reliability and observability platform**, improved data freshness, availability, and trust across large-scale **batch and streaming environments**.
- Designed and implemented **automated data quality and observability frameworks**, including freshness validation, schema drift detection, volume anomaly checks, and pipeline health monitoring.
- Developed **monitoring and alerting mechanisms** to proactively detect delayed, missing, or anomalous data, **reduced downstream business impact**.
- Integrated observability capabilities into **existing Spark and SQL-based pipelines**, enhanced reliability without compromising processing performance.
- Defined and aligned **data reliability SLAs with business expectations**, ensured measurable standards for **data availability, accuracy, and consistency**.
- Led **root cause analysis and incident resolution efforts**, correlated pipeline failures, data anomalies, and infrastructure signals to **reduce MTTD and improve system stability**.

IBM | Technology Services Specialist | Apr 2017 – Feb 2021

- Delivered **big data engineering and platform support services**, worked across Hadoop ecosystem components to enable **large-scale data processing and analytics capabilities**.
- Managed and supported **distributed data environments**, including cluster configuration, performance tuning, and system monitoring to ensure **high availability and reliability**.
- Developed and maintained **data processing workflow and ETL pipelines**, supported enterprise reporting and analytics use cases.
- Collaborated with cross-functional teams to **troubleshoot production issues**, optimize system performance, and ensure **SLA adherence**.
- Strengthened expertise in **data platform operations, cluster management, and performance optimization**, formed the foundation for advanced data engineering roles.
- Contributed to **secure data access and governance practices**, ensured compliance with enterprise data policies and standards.

Previous Experience

Future Focus Pvt. Ltd. (Deployed at TCS) | IT Analyst | Sep 2015 – Feb 2017

TCS | IT Analyst | Feb 2015 – Sep 2015

Genpact | Senior Associate | Jun 2011 – Feb 2015

Academic Details



2008
B.Sc. in Computer Science from
Sri Venkateswara University,
India



Personal Details

Languages Known: English, Hindi, Telugu, Tamil, Kannada Date of Birth: 2nd October 1988 Address: Bengaluru – 560016, India